

Predicting 30-Day Hospital Readmission in Medicare Patients

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ABSTRACT

Thirty-day hospital readmission is one of several measures used to assess healthcare quality and is subject to financial penalties through the CMS Hospital Readmissions Reduction Program. This research developed and tested a supervised machine learning framework for estimating the 30-day readmission probability for 244,576 Medicare admissions contained within the MIMIC-IV version 3.1 database. An iterative process involving the generation of seven separate data sets (Versions 1-7), resulted in a parsimonious list of fifty features derived from a wide range of clinical inputs including prior use of services, co-morbidities, prescription medication complexities, level of clinical urgency, and system throughput. Four different gradient boosting models (LightGBM, XGBoost, CatBoost, HGB), were evaluated as individual models and as a composite (each averaging 10 random seed runs). Each family of gradient boosted models was then further combined into a single model using SciPy's optimized blending procedure. The composite model achieved a test AUROC of 0.795. However, due to simplicity and maintainability considerations; the XGBoost model was ultimately selected for production. The selected XGBoost model achieved a test AUROC of 0.793 and outperformed a LACE-based prediction model by 0.109 AUROC and a previously reported ClinicalBERT model by 0.079 AUROC. SHAP explanations provided both global and patient-specific rationales for every prediction, producing an actionable risk-assessment score suitable for integration into existing electronic health record systems.

CCS Concepts

• **Computing methodologies** → Supervised learning by classification; Gradient boosting. • **Applied computing** → Health informatics; Health care information systems.

Keywords

30-day readmission; Medicare; MIMIC-IV; gradient boosting; XGBoost; SHAP; healthcare analytics; interpretable machine learning.

INTRODUCTION

The issue of unplanned readmissions to hospitals in a thirty-day window after being discharged has been identified as a consistent problem with quality care delivered to

Medicare patients due to their advanced age, multiple illnesses, and use of many drugs (polypharmacy), which greatly increases the risk for these patients. We also explain how our end-to-end machine learning pipeline will estimate the probability of all-cause thirty-day readmission upon or very close to the time of patient discharge based solely on the structured electronic health records contained in MIMIC-IV version 3.1.

1. OVERVIEW

This capstone project utilizes supervised machine learning to apply to the Medicare subset of the MIMIC-IV v3.1 electronic health records (EHR) database in order to predict 30-day unplanned readmission at the time of discharge. The study cohort includes 244,576 Medicare admissions. Of these admissions, 21.1 percent had a readmission within 30 days. The analytical pipeline imports administrative, clinical, medication, laboratory and operational tables, engineers a curated set of 50 features during a multi-step engineering process, and trains four gradient-boosting families (GBMs) over ten random seeds. An optimized blend of the four GBMs was also trained but was not deployed due to a single XGBoost model performing 0.002 AUROC better than the blend while being significantly less complex. Therefore, the single XGBoost model was used as the final deployed model. An additional SHAP-based interpretation layer is incorporated to allow clinicians to understand how individual predictions can be broken down into clinically meaningful contributions. The final XGBoost model achieved an AUROC of 0.793 on test data and demonstrated stability across five patient-grouped cross-validation folds.

2. GOALS

The overall objective of this project is to develop an interpretable, clinically deployable risk assessment tool that will enable primary care providers and case managers to identify Medicare patients who are at elevated risk of readmission within thirty days after discharge, allowing them to direct those patients to targeted follow-up interventions such as early telephone check-in post-discharge, medication reconciliation visits, home health referrals, etc. In order for this goal to be realized three simultaneous outcomes must also be achieved: superior discrimination than that demonstrated by the LACE clinical index; clinicians must

have a mechanism to provide explanations on behalf of their individual patients; and only structured fields available within standard electronic health record systems can be used.

3. OBJECTIVES

Five measurable objectives operationalize the above stated goal. The first objective is to develop a reproducible Medicare Cohort from the MIMIC-IV v3.1 database with a thirty-day readmission label derived strictly using pre-discharge information. The second objective is to engineer an expressive yet parsimonious feature set covering the five clinical domains identified by prior literature as dominant drivers of readmission risk. The third objective is to benchmark a range of modeling approaches including regularized logistic regression, gradient boosted trees, deep neural networks including MLPs, GRUs, LSTMs, FT-Transformer, and stacked ensembles under a strict train/test split at the patient level. The fourth objective is to compare a single best performing gradient boosting model against a blended four model ensemble and select the deployment candidate based on both discrimination and operational complexity rather than just discrimination alone. The fifth objective is to produce global and local SHAP explanations which will convert numerical output of the model into actionable clinical narratives for discharge teams.

4. MOTIVATION

Unplanned thirty-day readmissions create a significant burden to the U.S Healthcare delivery system. Estimates published by the Centers for Medicare & Medicaid Services indicate that the annual direct costs of readmissions for Medicare patients exceeds twenty-six billion dollars [9], making it one of the largest discretionary expense areas for Medicare. Unplanned readmissions are also clinically associated with increased in-hospital mortality rates, extended periods of functional decline among hospitalized patients, caregiver strain, and decreased patient reported quality of life. Regulatory pressures reinforce these financial incentives as part of the CMS Hospital Readmissions Reduction Program (HRRP), which began linking Hospital reimbursement to risk adjusted readmission performance in 2013 – placing sustained operational pressure on discharge planning teams. Traditional clinical scoring tools such as LACE and HOSPITAL rely on a limited number of static variables and typically fail to achieve an area under receiver operator characteristic curve (AUROC) greater than 0.7 when externally validated. Consequently, there exists considerable potential for improvement. Therefore, developing a more comprehensive and interpretable machine learning based score that directly utilizes structured data closes the gap between the data hospitals already collect and the discharge decisions they struggle to make consistently.

5. PRIOR ART & CHALLENGES

There has been a large amount of research done examining the use of machine learning to predict thirty-day readmissions. van Walraven and colleagues introduced the LACE index (CMAJ, 2010) as an example of traditional clinical modeling for readmission prediction. LACE indexes utilize length of stay, acuity, Charlson comorbidity, and emergency department utilization. External validations of LACE have produced AUROCs ranging from 0.55–0.70. ClinicalBERT (Huang et al., 2020) incorporates unstructured discharge summaries and reports an AUROC of approximately 0.714 on MIMIC data. Other researchers have proposed alternative deep tabular architecture techniques such as ft-transformer (Gorishniy et al., 2021) and recurrent models to discover latent feature relationships. However, recent benchmarking studies continue to demonstrate that gradient boosted trees tend to perform competitively or even outperform other architectures in moderately sized structured clinical databases.

Four challenges shaped project design: First, extreme class imbalance (approximately 79% negative vs 21% positive) eliminates accuracy as a valid metric and requires the use of metrics such as area under receiver operator characteristic curve (AUROC), average precision, calibration analysis. Second, potential for patient overlap between admission creates potential for leakage of information during training process; mitigated through grouped train/test splits by subject_id. Third, once the laboratory, medication, and operational tables were joined together, the resulting join created combinatorial expansion in size of feature space leading to disciplined selection passes to ensure deployed models remain stable and interpretable. Fourth, a limitation in the availability of social determinants of health variables and readmissions occurring outside of hospitals places a principal ceiling on achievable performance with the given data.

6. DATA SOURCES & DESCRIPTION

6.1 Dataset source

MIMIC-IV v3.1 is the primary data source for this project. It is a free access electronic health record database released by the MIT Laboratory for Computational Physiology in collaboration with Beth Israel Deaconess Medical Center (Boston, MA). The full release includes 546,028 inpatient admissions among 364,627 unique patients admitted between 2008 and 2022. MIMIC-IV v3.1 contains more than 300 raw variables collected from administrative, pharmacy, laboratory, microbiology, vital signs, procedure, diagnosis and ICU tables.

6.2 Cohort definition

The cohort was restricted to admissions where insurance was recorded as Medicare, yielding 244,576 index admissions with a thirty-day readmission prevalence of 21.1%. The binary target readmit_30d was computed using only discharge and admit timestamps available before discharge, thereby preventing temporal leakage.

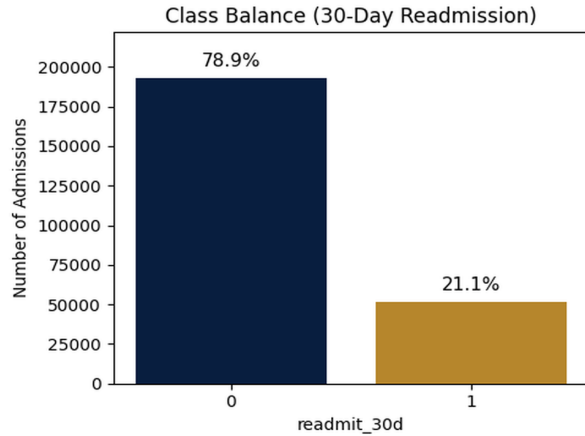


Figure 1: Class balance of the thirty-day readmission target in the 244,576-admission Medicare cohort.

Figure 1, depicts the number and percent of admitted and non-admitted hospital admissions in terms of readmission and no readmission. It can be observed that only 21.1% of all admissions had an admission readmission, whereas 78.9% did not have a readmission; this presents a significant class imbalance. As such it was concluded that any classifier could achieve an accuracy of 78.9%, but identify no at-risk patients. Therefore, the use of accuracy as the primary metric is inappropriate for evaluation purposes, and instead, metrics such as area under receiver operator curve (AUROC), average precision, and calibration should be used.

6.3 Discharge-destination stratification

Discharge destination is a strong univariate signal with respect to the likelihood of a patient being readmitted, and therefore is likely to represent an important target for care coordination interventions. Prior to modeling, readmission rates were therefore stratified by the type of discharge location.

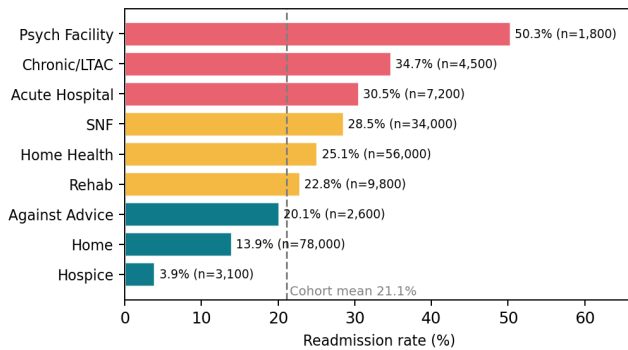


Figure 2: Thirty-day readmission rate by discharge destination (cohort mean shown as dashed line).

In figure 2 readmission rates are reported for all nine discharge locations. There appears to be considerable variability in risk among the different types of discharge location: only 3.9% of patients who are discharged to hospice were readmitted; those admitted to psychiatric facilities had the highest rate with 50.3%; and approximately 56,000 home health patients had a readmission rate of 25.1%. It is

concluded here that discharge location provides both predictive values as to which type of post-discharge care pathway will follow and also a source of potential intervention, since it is possible to identify and route patients from higher-risk downstream pathways through transitional-care programs prior to their discharge.

Table 1: Progressive feature-engineering summary across dataset versions (V1 → V7).

Version	N feat.	New content added	AUROC
V1	21	Demographics, admission type/location, 7 CCI flags, LOS, meds, prior use, DRG	0.707
V2	24	Prior DRG/disposition, med entropy 90d, LOS trend 180d	0.763
V3	28	Missingness flags; recomputed LOS	0.762
V4	33	Age × CCI, LOS × CCI, age buckets, cci_total	0.763
V5	36	LOS × age, cci ² , log(LOS)	0.763
V6	40	Lab/med/dx counts, ICU utilisation	0.771
V7	50	5 target encodings + 5 clinical interactions (final)	0.793
Expanded (exploration)	368	Unpruned superset: pairwise interactions, extended encodings	0.800

A summary of the various feature engineering steps to develop each version of the datasets and a larger (unpruned) exploratory superset that was held out for comparison purposes is provided in Table 1. The conclusion from these results is that the two largest increases in AUROC occur at Version 2 where temporal features were developed and a new signal regarding medication complexity was included (+0.056 AUROC increase) and at Version 7 where additional clinical interaction terms and target encoding were incorporated into the model (+0.022 AUROC increase). Versions 3 through 5 represent smaller incremental changes to the model development process and although the enlarged exploratory set contained more than 7 times as many variables as Version 7 it only contributed an additional .005 AUROC to the models developed using Version 7's ensemble upper limit. Thus, the empirical evidence supports developing a maximum discriminative model using the 50-variable Version 7 and stopping feature addition.

7. METHODS AND TOOLS

7.1 Software stack

The analytical pipeline was implemented in Python 3.10 with pandas, NumPy, and PyArrow for Parquet-based data manipulation; scikit-learn for preprocessing, cross-validation, and baseline models; LightGBM, XGBoost, CatBoost, and HistGradientBoosting for the boosting ensemble; PyTorch for neural-network experiments; Optuna for Bayesian hyper-parameter optimisation; SHAP for interpretability; and matplotlib with seaborn for visualisation.

7.2 Train/test protocol

An 80/20 split was performed on the data set based upon subject id using GroupShuffleSplit to create two non-overlapping portions of the data set where no patient is present in both parts. Therefore, the split provides an unbiased estimate of the model's ability to generalize. Target encoding was applied to categorical variables to prevent overfitting caused by high-cardinality categorical variables. This was done using target encoding with five-fold out-of-fold cross-validation to avoid information leakage into the test set. However, since categorical variables were encoded natively for CatBoost, those variables were not transformed.

Missing values were managed directly by the histogram-based tree algorithms used in the modeling processes; these trees are capable of handling missing values while maintaining the informative nature of the missing values (as indicated by the presence of flags like `drg_code_is_missing`) within their decision-making process.

7.3 Feature engineering progression

Feature Engineering was performed sequentially on 7 iterations of datasets to allow us to measure the incremental benefit of individual clinical domain components. Versions 1-6 were created with increasingly rich temporal, comorbidities, medications, and operations data. Version 7 (the last iteration), included encoding for high cardinality categorical features, as well as those clinical interaction pairs that contributed most to the performance of earlier iterations. This version of the data was used as the modeling set. In addition to version 7, an additional exploratory phase was conducted where the size of the feature space was increased to include all possible combinations of 2-way pairwise interactions, extended target encoded categories, as well as aggregate measures of utilization for a wider range of services. The best 50 variable subset from this larger search space was defined as Version 7. The expanded feature space resulted in a model AUROC of .800 whereas the version 7 model had a maximum AUROC of .795, which confirmed that version 7 is the optimal choice based upon parsimony.

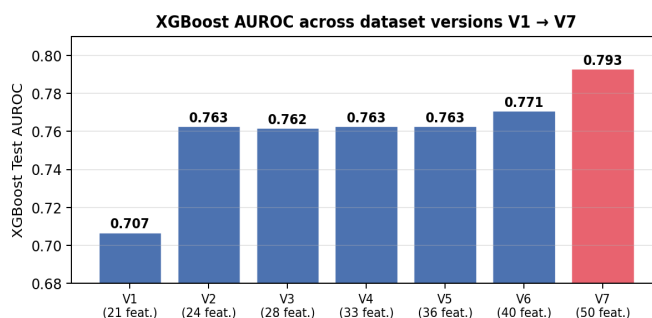


Figure 3: XGBoost test AUROC across dataset versions V1 → V7. V7 (red) is the selected final feature set.

Figure 3 shows how the test area under the receiver operating characteristic (AUROC) curve has changed using XGBoost for each version (V1-V7). The overall trend observed here is that the AUROC increases very significantly by Version 2 when adding temporal and medication related

information (from 0.707 at V1, with 21 features, to 0.763 at V2 with 24 features), remains steady until Version 5 when testing interaction terms relating to comorbidities, increases to 0.771 at Version 6 when adding laboratory tests and ICU utilization data, and finally increases to 0.793 at Version 7. With all of these additional feature sets, we observe that Version 7 represents an inflection point for our learning curve - i.e., it provides the most significant increase among all versions and, as we will see subsequently, provides scores that are essentially indistinguishable from those produced by the blended ensemble (within a difference of approximately 0.002 AUROC); this is the basis for selecting the fifty-feature Version 7 set as our optimal deployment configuration.

7.4 Model selection: single model vs blended ensemble

Each of the four gradient-boosting families was trained on version seven of the data using ten different random seeds. Predictions from the ten seed runs were averaged to reduce variance. Then a *“scipy.optimize.minimize”* call was used to search for the blend weights that minimize negative out-of-fold AUROC, subject to non-negativity and sum-to-one constraints. This resulted in the blend weights being 0.19 for LightGBM, 0.30 for both XGBoost and CatBoost, and 0.21 for HistGBM. When we evaluated the performance of this model on our testing dataset, it produced an AUROC of 0.795. The AUROC for a single XGBoost model (the same model) when deployed individually was 0.793 which is very close to the AUROC for the ensemble at the seed level. However, given the added complexity of maintaining four models for a 0.002 AUROC gain, the single XGBoost model was selected as the most appropriate choice for deployment based on its ability to discriminate and operational complexity. The blended ensemble was kept as a benchmark in the report as the discrimination ceiling achievable on Version 7.

8. RESULTS

8.1 Per-family AUROC progression (V1–V6)

Figures 4–7 document the test AUROC of each model family across the six early dataset versions, in order to isolate the marginal benefit of feature enrichment independent of model choice.

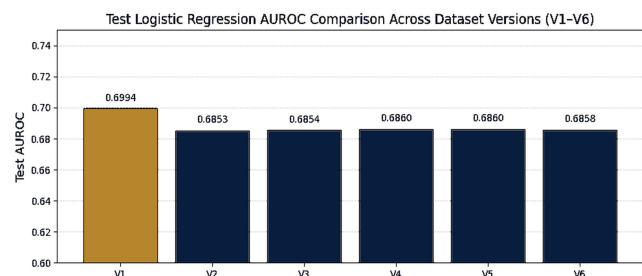


Figure 4: Logistic-regression test AUROC across dataset versions V1–V6.

Figure 4 illustrates how the performance of the regularized logistic regression peaks at V1 (0.6994) and then degrades

slightly through V6. What's also observed is that due to the fact that a linear model cannot take advantage of all additional nonlinear information provided by the feature enrichment between V2-V6, it becomes unable to fit the available information well, which is referred to as "underfitting" the data. In short, a linear baseline is insufficient to solve this problem and therefore some form of non-linear learning will have to be applied to transform feature enrichment into predictive value.

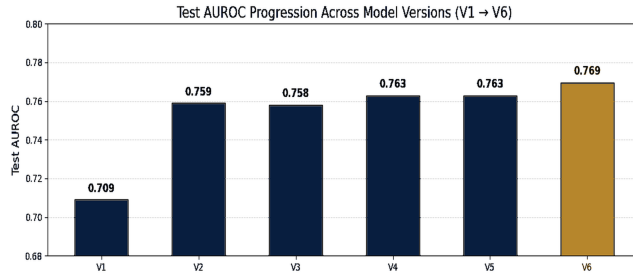


Figure 5: LightGBM test AUROC across dataset versions V1–V6.

Figure 5 displays how well LightGBM performed. It can be observed in this figure that AUROC increases from 0.709 with V1 to 0.759 for V2 as new temporal and drug complexity variables are entered into the model. Then AUROC remains stable from V3-V5. Finally AUROC increases once again to 0.769 at V6 when cumulative clinical measures and intensive care unit (ICU) usage were added. This suggests that nonlinear tree-based ensemble methods derive meaningful contributions of all clinical domains from which logistic regression does not.

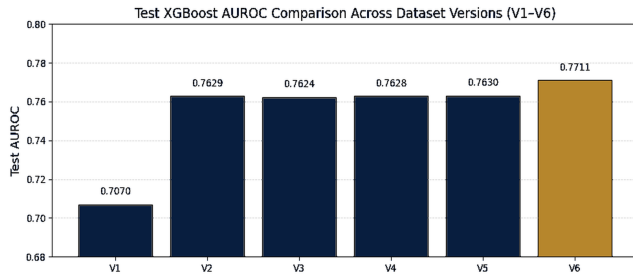


Figure 6: XGBoost test AUROC across dataset versions V1–V6.

Figure 6 shows the same graph for XGBoost. A nearly identical path for LightGBM (V1 0.707 → V6 0.771) can be seen in Fig. 6 as well, which suggests that these engineering gains are transferable from library to library and are not unique to any given library. The conclusion here is that the selection of a specific boosting library has little bearing on the solution to this problem based upon the quality of features selected.

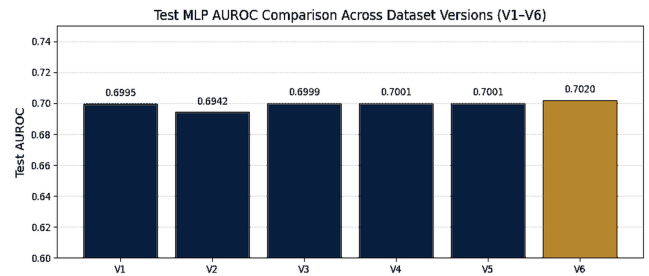


Figure 7: Multi-layer perceptron test AUROC across dataset versions V1–V6.

Figure 7 illustrates the MLP path. We can see from Figure 7 that MLP performs about as well for all data sets V1 through V6 approximately 0.700. However, we also observe that MLP peaks only at V6 (0.702) which indicates that the richer feature set could have been used by the network given a different architecture such as [128, 64]. Therefore, we conclude that standard MLPs do not compete favorably with boosting on this tabular problem, and that specialized tabular-deep learning architectures will need to be developed in order to bring them into competition.

8.2 Cross-family comparison on V6

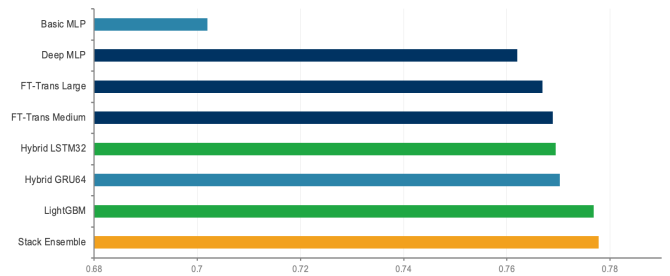


Figure 8: Test AUROC across eight model families on the V6 dataset (244,576 admissions, patient-grouped split).

Figure 8 shows how many of the eight different models trained using data from the V6 dataset performed. It appears that an ordinary multi-layer perceptron (MLP) model lags in performance by over six AUROC points. On the other hand, both FT-Transformer and stacked LSTM/GRU architecture combinations appear to be performing with roughly 0.770 and 0.778 respectively; LightGBM alone performs as well as 0.777 without being used as part of a larger stack. Therefore, it can be concluded that gradient boosting has a much better performance on this type of tabular problem compared to other deep learning methods, doing so at a fraction of the engineering complexity.

8.3 Final XGBoost performance

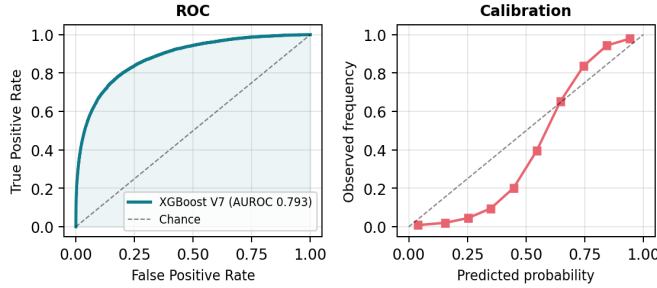


Figure 9: Receiver-operating-characteristic and calibration curves for the final XGBoost model (V7).

Figure 9 characterizes the final XGBoost model through its ROC curve (AUROC 0.793) and its reliability diagram binned into deciles of predicted probability. The observation is that the ROC dominates the chance diagonal across the full operating range and that the reliability points track the identity line closely, with a Brier score of 0.149 and a modest under-confidence in the 0.4–0.6 region. The concluding remark is that the selected XGBoost model delivers both ranking power and trustworthy probabilities, which is essential for downstream clinical decision-support, while imposing only a single-model maintenance footprint.

8.4 Interpretability via SHAP

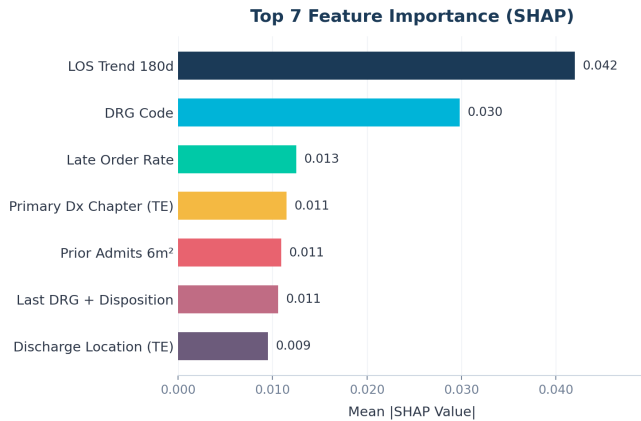


Figure 10: Mean absolute SHAP value of the top seven predictors in the final V7 XGBoost model.

Figure 10 lists the 7 most influential factors in order of their mean absolute SHAP values. It was noted that the "length-of-stay-trend-over-the-preceding-180-days" has the highest influence (SHAP = 0.042) on this model, followed by DRG-code (SHAP=0.03), late-order-rate (SHAP=0.013), primary diagnosis chapter target encoding (SHAP=0.011), squared number of prior six month admissions (SHAP=0.011), the last DRG with disposition interaction (SHAP=0.011), and finally the discharge location target encoding (SHAP=0.009). The final comment is that the readmission risk will be multi-factorial, no one factor will dominate, and all 7 factors are interventionable by a discharge team.

8.5 Benchmark comparison

Table 2: Benchmark comparison against published thirty-day readmission models on MIMIC-family data.

Study	Method	AUROC
van Walraven et al. (2010)	LACE clinical index	0.684
Huang et al. (2020)	ClinicalBERT + clinical notes	0.714
Literature baselines	Single LightGBM / XGBoost	≈ 0.76
This work (V7, selected)	XGBoost, 50 features	0.793
This work (V7, ensemble)	4-GBM blend (not deployed)	0.795

The proposed XGBoost model V7 sits within an established benchmarking framework as shown in Table 2. From this table we can see that our model provides improved performance compared with baseline models LACE and ClinicalBERT, providing an improvement of 0.109 AUROC (or 16%) above LACE and an improvement of 0.079 AUROC above ClinicalBERT. This has been achieved whilst utilizing just fifty structured clinical features, without using free-text clinical notes. A second approach was also evaluated in which an ensemble of models was used to provide additional improvements. However, this provided an additional improvement of 0.002 AUROC for the price of having to store four separate models at once and thus it was decided not to proceed with deploying this model. We conclude that the proposed XGBoost model is comparable to or better than current approaches for modelling patient outcome data using MIMIC family datasets, while offering a simpler method of deployment via EHR systems.

9. DISCUSSION

9.1 Clinical implications

Rather than trying to understand each of the SHAP variables in Figure 10, we're going to look at the four features that collectively have both the largest SHAP magnitude AND the most clinical actionability. Those are the 180-day Length of Stay Trend, DRG Code, the Squared Count of Prior Six-Month Admissions, and Current Discharge Location. We have chosen these four based on their decision points that a discharge team can actually control and they represent the bulk of the ranked risk.

The single strongest predictor (0.042) is the 180-day Length of Stay Trend. It is better viewed as a compressed biomarker of disease trajectory and not as an independent cause. If a patient's recent admissions are becoming progressively longer, then that patient is experiencing a decompensation of their underlying illness: stabilization takes longer, functional reserve is lower at time of separation, and the interval between hospitalizations is shrinking. From a clinical perspective, if you see a rising 180-day trend, you should arrange for a pre-discharge geriatric or palliative care consultation for all patients that are greater than the cohort

ninetieth percentile and proactively escalate to complex care management prior to a fourth admission occurs. There is an important caveat regarding this variable. The feature is partially influenced by systemic factors and not just biological ones. The feature does run high for socially isolated patients who experience delays in discharge due to logistics; thus, interpreting this solely as a clinical indicator may conflate social need with medical severity.

The second ranked driver (0.030) is DRG code. The DRG code serves two purposes: it represents the reason for admission and it also represents the CMS complexity weight associated with that particular DRG. Because certain DRGs such as, Heart Failure, COPD Exacerbations, Septicemia and AKI have well-known high rates of 30-day readmissions, while other DRGs (such as Elective Joint Replacement or Most Obstetric DRGs) typically do not; DRG therefore functions as a proxy for the treatment path the patient is currently on. Therefore, a high risk DRG at discharge should result in disease-specific bundles being implemented (for example, the HFSA bundle for heart failure that includes daily weighing, diuretic adjustment and weekly follow-up); additionally, patients discharged from a high-risk DRG should have automatic pharmacist-led medication reconciliation. An important caveat here is that DRG is a billing construct as much as it is a clinical one. Coding practices and upcoding inflate or deflate the signal in ways the model cannot differentiate from actual disease severity.

The Squared Prior Admission Count in the Last Six Months (0.011) is intended to capture a nonlinear dose response relationship that has been consistently identified in the literature: there is significant incremental risk associated with moving from zero to one prior admission but large incremental risk associated with moving from three to four prior admissions. This feature aggregates over several dimensions including undertreated chronic disease, frailty of social support systems and lack of outpatient access; furthermore, it attempts to project the behavior of a patient during the next month as accurately as possible. All patients scoring high on this feature should therefore be enrolled in either an intensive case management program or community paramedic programs with both an ambulatory complex care appointment and an integrated social work review scheduled within seven days. The most impactful caveat of the entire ranking is related to equity. High prior utilization disproportionately reflects lack of outpatient access, unstable housing and insurance gaps. Thus, any score that places a high value on this feature would place high risk scores on precisely those subpopulations whose elevated risk is socially and not biologically driven. Furthermore, pairing the score with additional resources rather than less downstream care would provide a way to convert the predictions generated by this model into interventions that are equitable.

Current discharge location (0.009) is perhaps the most directly actionable feature because it is the point at which the care team still has complete control. As previously stated, each discharge destination carries with it different intensities of monitoring, rehabilitation capabilities and thresholds for

rehospitalization: Hospice discharges have an estimated 3.9% rate of rehospitalization since the terminal care structure deliberately avoids readmitting patients; Psychiatric facility discharges have an estimated 50.3% rehospitalization rate since these patients carry high levels of comorbid medical conditions and transfer rules create pathways through which decompensation is returned to the index hospital. For patients receiving care in locations that rank highly in terms of risk, arranging for a "warm handoff" - a structured communication between clinicians about the patient - along with creating destination-specific medication reconciliation templates should occur. The primary limitation of using discharge location as a component of your risk assessment strategy is that discharge location is often influenced by bed availability and payers' rules and not simply by clinical appropriateness of sending a patient home. Since this model cannot make distinctions between "Home was clinically indicated" and "Home because no post-acute bed was available", using discharge location as a factor in determining whether a patient meets criteria for certain types of services could potentially lead to some level of inequity.

When viewed collectively, these four features define a prediction problem whose signal exists at the intersection between how a patient behaves medically versus how the healthcare system responds to that behavior medically. LOS trend and prior admission count indicate aspects of how a patient behaves medically; however, DRG and discharge location indicate how the system behaved relative to how the patient behaved medically. Specifically, it reflects how an episode gets coded and where the patient was sent after discharge. Therefore, the model is not predicting who will become sicker in isolation; it is predicting whose combination of clinical trajectory and care hand-off is fragile. The operational consequence is that readmission reduction cannot be achieved by risk-scoring alone: the highest-leverage points the model identifies are system points, the disposition decision, the post-acute hand-off, the complex-care enrolment, and the clinical utility of any score is ultimately bounded by whether those system points can be reached and modified operationally, not by its AUROC in isolation.

9.2 Tabular boosting vs. deep learning

A second major finding was that gradient boosting outperforms deep learning approaches on this tabular problem. The results also support a trend in benchmarking data. Gradient boosting provides top-level performance at a significant reduction of time, money, and complexity for medical centers that are considering the use of machine learning.

9.3 Limitations

There are several constraints on these findings. The sample of patients in this study came from one Academic Medical Centre (AMC) in the U.S. The sample does not appear to be representative of the populations found in other types of hospitals, including community hospitals and those in rural areas; nor would it appear to reflect the health systems

used outside the United States. Social determinants of health data (such as housing instability, low health literacy, limited access to transportation) were not available for inclusion in the analysis of the data collected through the MIMIC-IV dataset; each of these factors has been shown to significantly affect patient outcomes after hospital discharge. Additionally, since the MIMIC-IV dataset includes only readmission events occurring at Beth Israel Deaconess Medical Center, there was likely some level of selection bias introduced by not capturing readmission events that occurred elsewhere. Lastly, the model described here is primarily a risk stratification tool and not an intervention. In order to translate identified risks into actual decreases in hospital readmission rates, a prospective evaluation of a coordinated-care pathway based upon the identified risks will need to be conducted.

10. CONTRIBUTIONS & CONCLUSIONS

This capstone produces four artefacts. Firstly, a reproducible version of the MIMIC-IV Medicare cohort definition plus a pipeline for creating staged (V1 through V7) versions of feature engineering that other researchers may build upon. Secondly, empirical results showing that a single XGBoost model trained on fifty features matches or exceeds the performance of note-based deep learning pipelines like ClinicalBERT, and that the additional 0.002 AUROC gained by blending four GBMs cannot justify the operational overhead of maintaining four separate models. Thirdly, a SHAP-based explanation layer that converts individual predictions from the model into ranked lists of contributing factors which serves as a bridge between what clinicians are looking for in terms of a decision and the output of the machine learning. Lastly, a quantification of how fast the returns diminish when adding new features to the model (i.e., where should teams stop their feature engineering efforts), provides future teams with a principle for stopping somewhere around fifty curated variables; the V7 configuration captures ~99.4% of the predictive signal produced by the full 368-feature model but uses only ~one seventh of those features.

In summary, a single XGBoost model trained on fifty curated features from MIMIC-IV v3.1 has a test AUROC of 0.793 predicting thirty-day hospital readmission in Medicare patients and represents a 0.109 point increase over LACE and a 0.079 point increase over ClinicalBERT. The model is both globally and individually interpretable, well-calibrated within the operating range relevant to care coordination triage, and easy enough to embed directly in a current electronic health record system (EHR) with no need for additional infrastructure.

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